1.Use Figma to create an interactive diagram illustrating the components of a basic cellular system.

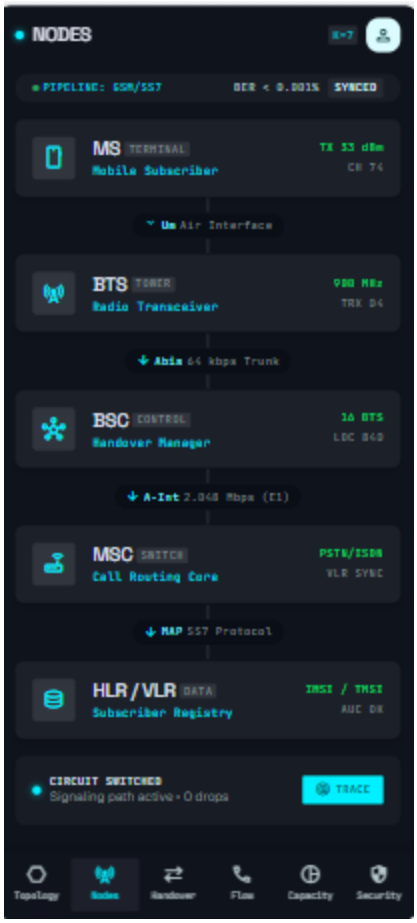
Aim: -

To create an interactive diagram illustrating the components of a basic cellular system

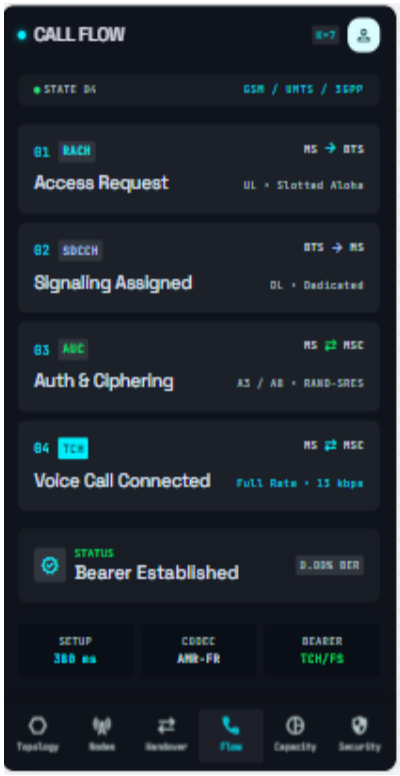
Procedure: -

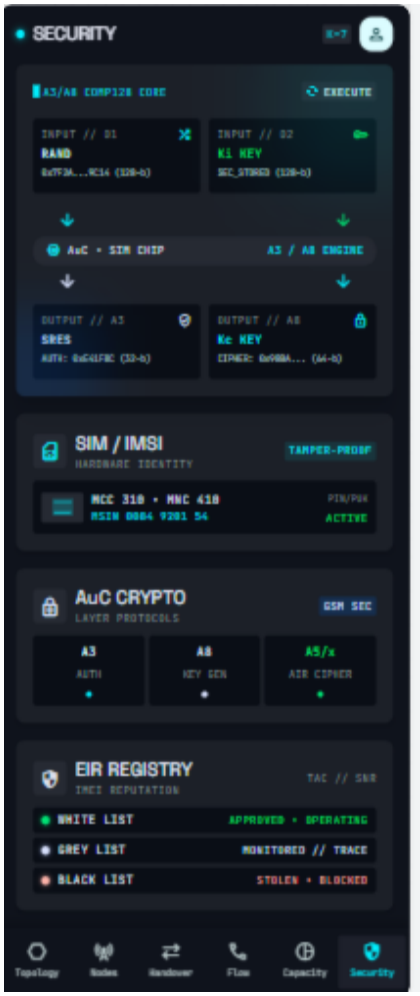
1. open Figma
2. Create a new file
3. Select the Frames
4. Fill in the content that is required for presentation 5. Design Visual Elements
6. Make it Interactive
7. Add Annotations and Explanations
8. Incorporate Multimedia
9. Storyboard Animation
11. Review and edit the Prototype
12. Save and Share

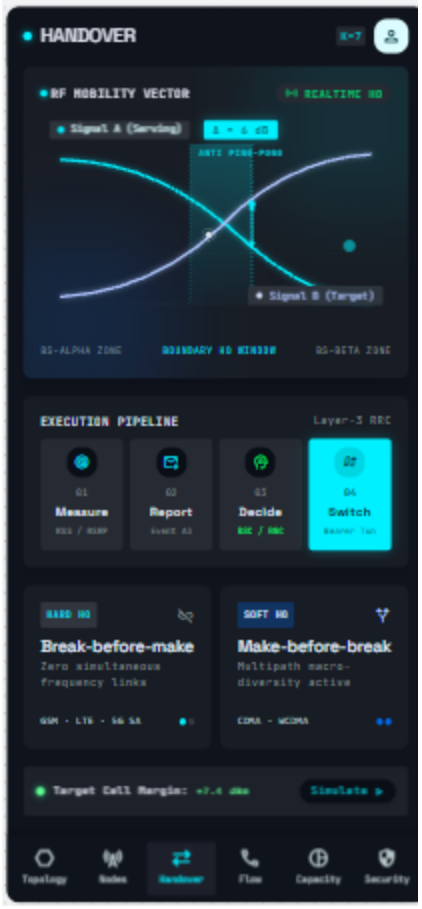
Design: -

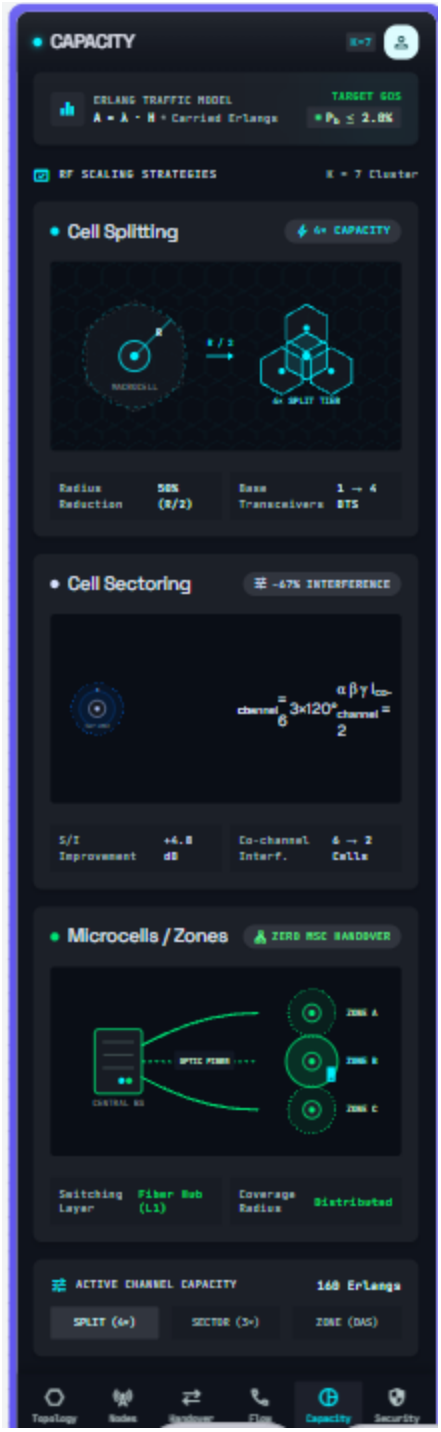

NODES
N=7
PIPELINE: GSM/SS7
BER < 0.001%
SYNCED
MS TERMINAL
Mobile Subscriber
TX 33 dBm
CH 74
Um Air Interface
BTS TOWER
Radio Transceiver
900 MHz
TRX 04
Abis 64 kbps Trunk
BSC CONTROL
Handover Manager
16 BTS
LOC 040
A-Int 2.048 Mbps (E1)
MSC SWITCH
Call Routing Core
PSTN/ISDN
VLR SYNC
MAP SS7 Protocol
HLR / VLR DATA
Subscriber Registry
IMSI / TMSI
AUC OK
CIRCUIT SWITCHED
Signaling path active · 0 drops
TRACE
Topology
Nodes
Handover
Flow
Capacity
Security


TOPOLOGY
RF CLUSTER // GEOM
7-CELL PATTERN
N = 7
CLUSTER SIZE
D = 4.6R
REUSE DIST
≥18 dB
S/I TARGET
FREQUENCY REUSE
Same RF channel repeated without overlap
CO-CHANNEL
Spatial distance prevents signal clash
Topology
Nodes
Handover
Flow
Capacity
Security


CALL FLOW
STATE 04
GSM / UMTS / 3GPP
01 RACH
MS → BTS
Access Request
UL · Slotted Aloha
02 SDCCH
BTS → MS
Signaling Assigned
DL · Dedicated
03 A&C
MS ⇄ MSC
Auth & Ciphering
A5 / A8 · RAND-SRES
04 TCH
MS ⇄ MSC
Voice Call Connected
Full Rate · 13 kbps
STATUS
Bearer Established
0.00% BER
SETUP
380 ms
CODEC
AMR-FR
BEARER
TCH/FS
Topology
Nodes
Handover
Flow
Capacity
Security


SECURITY
A3/A8 COMP128 CORE
EXECUTE
INPUT // 01
RAND
INPUT // 02
Ki KEY
AuC • SIM CHIP
A3 / A8 ENGINE
OUTPUT // A3
SRES
OUTPUT // A8
Kc KEY
SIM / IMSI
HARDWARE IDENTITY
TAMPER-PROOF
MCC 310 • MNC 410
MSIN 0004 9201 54
ACTIVE
AuC CRYPTO
LAYER PROTOCOLS
GSM SEC
A3
AUTH
A8
KEY GEN
A5/x
AIR CIPHER
EIR REGISTRY
IMEI REPUTATION
TAC // SNR
WHITE LIST
APPROVED • OPERATING
GREY LIST
MONITORED // TRACE
BLACK LIST
STOLEN • BLOCKED
Topology
Nodes
Handover
Flow
Capacity
Security


HANDOVER
RF MOBILITY VECTOR
REALTIME HO
Signal A (Serving)
ANTI PING-PONG
Signal B (Target)
EXECUTION PIPELINE
Layer-3 RRC
01
Measure
02
Report
03
Decide
04
Switch
HARD HO
Break-before-make
Zero simultaneous frequency links
SOFT HO
Make-before-break
Multipath macro-diversity active
Target Cell Margin:
Simulate
Topology
Nodes
Handover
Flow
Capacity
Security


CAPACITY
K=7
ERLANG TRAFFIC MODEL
A = λ · H · Carried Erlangs
TARGET GOS
RF SCALING STRATEGIES
K = 7 Cluster
Cell Splitting
6× CAPACITY
R / 2
6× SPLIT TIER
Radius Reduction
50% (R/2)
Base Transceivers
1 → 4 BTS
Cell Sectoring
-67% INTERFERENCE
S/I Improvement
+4.8 dB
Co-channel Interf.
6 → 2 Cells
Microcells / Zones
ZERO MSC HANDOVER
OPTIC FIBER
CENTRAL BS
ZONE A
ZONE B
ZONE C
Switching Layer
Fiber Hub (L1)
Coverage Radius
Distributed
ACTIVE CHANNEL CAPACITY
160 Erlangs
SPLIT (6×)
SECTOR (3×)
ZONE (DAS)
Topology
Nodes
Handover
Flow
Capacity
Security

Prototype: